

Syllabus

Spring semester 2025

Course Title	Introduction to Nanochemistry		Course Code		CB4103		Credit	3	Class room	Hyung 3303 & Inorganic Lab
			Type		elective		Hours	3		
Teacher	Name		Class no.	Office	Office hours		Contact		E-mail	
	Choi, Eun-Young		1	chang603	M8, Tu8, F 7		606-2389		faujasite1@kaist.ac.kr	

1. Course Description

This course is for learning to types and applications of nano solid materials, such as activated carbons, metal organic frameworks, zeolites, ceramics, polymers etc, attracting interests. Understand the recent trends of nano materials that can be used as gas storage and catalysis and recognize the needs of development of nano materials. Learn the basics of Inorganic Chemistry to understand properties and applications of nanosolid materials. Stimulate interests and concerns, for promising nano materials and learn how to get the proper information and understand them with various tools such as database, chemdraw, visualizer program.

2. Core Target Competencies

Subject Core Competencies						
Analytical and Creative Thinking ¹⁾	Complex Communication ²⁾	Leadership and Teamwork ³⁾				
1) Ability to find quality information in the flood of information to articulate meaningful questions, to analyze and create ideas or knowledge 2) Ability to communicate clearly with diverse audiences or readers, to communicate clearly with diverse audiences or readers, to listen attentively 3) Ability to be the first to come up with and present new ideas, tasks, manage groups, and delegate responsibilities						

3. Textbooks and References

A. Textbook: Workbook (2025 Introduction to Nanochemistry) and Selected Papers

B. References: Descriptive Inorganic, Coordination, and Solid-State Chemistry, ACS papers (Zeolites, Metal Organic Framework, Coordination Complex, Coordination polymers)

4. Class Method

- A. Lecture
 - A Class that stimulates active student participation
- B. Debate/Discussion
- C. Activities
 - Practical exercises utilizing programming and database applications
 - Nano chemistry Experiment
 - Molecular model kit

5. Evaluation

A. Evaluation Criteria

Mid-term	Project Presentation	Experiment & Report	Attendance	Attitude	Total*
35	25	20	10	10	100%

B. Evaluation Method

- (1) The grade will be given based on total points.
- (2) Peer review will be included in the grading criteria.
- (3) Attendance and Attitude
 - No absences and No late arrivals Full score (10 points)
 - Being late once: -0.2 points.
 - One, two, and three absence: -1, -2, and -5
 - Students with more than four unexcused, full-class absences will receive an "F" grade for the course.
 - Using electric device without permission (e.g., chatting on line, playing games): -10 points.

6. Criteria for Awarding Credits

☛ Absolute Evaluation

- ☛ Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.

7. Weekly Plans

Week (period)	Topic and Contents	(Notes)
Week 1 2/17-2/21	Orientation Introduction to nano-porous materials, MOF & Zeolite	Lecture
Core Competencies	♦ Analytical and Creative thinking, Leadership and Teamwork	
Week 2 2/24~2/28	Crystal geometry1 - Crystal axis, Unit cell, Interstitial sites, Classification of Crystals	Homework: p 8
Core Competencies	♦ Analytical and Creative thinking, Leadership and Teamwork	
Week 3 3/3~3/7	Crystal geometry2 - Miller index, Problems	Homework p21
Core Competencies	♦ Analytical and Creative thinking, Complex communication	

Week 4 3/10~3/14	Understanding symmetry for Nano-Inorganic Solids - Symmetry element, Point group	Lecture and activity
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Leadership and Teamwork	
Week 5 3/17~3/21	Coordination chemistry, nomenclature, complex, coordination polymer	Lecture and activity
Core Competencies	♦ Analytical and Creative thinking, Complex communication	
Week 6 3/24~3/28	Understanding of Coordination Chemistry, Metal Organic Framework	Lecture, Activity
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Leadership and Teamwork	
Week 7 3/31~4/4	Understanding of zeolites	Lecture, Activity
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Leadership and Teamwork	
Week 8 4/7~4/11	Midterm Exams	
Week 9 4/14~4/18	Solvothermal Synthesis of MOF-5	Experiment worksheet
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Leadership and Teamwork	
Week 10 4/21~4/25	Observation Crystals and Create Image Data	Lecture, Activity
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Leadership and Teamwork	worksheet
Week 11 4/28~5/2	Understanding MOF Structures using visualizer program, mercury Metal geometry: CSD (Cambridge Structural Database):	Lecture, Activity
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Leadership and Teamwork	worksheet
Week 12 5/5~5/9	-Understanding of Nanoporous materials - making a metal cluster and building up framework of nano-porous materials using <u>Molecular model kit</u>	Activity
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Leadership and Teamwork	
Week 13 5/12~5/16	Paper summary presentation	Presentation
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Leadership and Teamwork	
Week 14 5/19~5/23	Paper summary presentation	Presentation
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Leadership and Teamwork	
Week 15 5/26~5/30	Inorganic crystalline materials: zeolites - Stereoviews of crystal structures, magic eye	Lecture, Activity
Core Competencies	♦ Analytical and Creative thinking, Complex communication, Leadership and Teamwork	
Week 17 6/9~6/13	Final Exams	

※ The above plans may change according to the academic schedule.

[Reference] Syllabus Competence Reference

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/ Quantitative Literacy	5. Global Perspective	6. Adaptability/ Progression / Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/ Mindset
Ability to identify, manage and deal with complex problems	Ability to understand and express themselves in two or more languages	Ability to be the first to come up with and present new ideas	Ability to understand, use and apply digital technology	Ability to develop an open mind about the traditions or values of others	Ability to develop flexibility, agility and adaptability	Ability to sustain an empathetic and understanding perspective	Sincerity
Ability to see through prejudice and identify reliable information and uncertain information	Ability to communicate clearly with diverse audiences or readers	Ability to influence people to follow themselves, rather than lead them through directives, orders, or authority	Ability to create digital contents and media	Ability to understand the history, politics, religion, and culture of another region	Ability to challenge unfamiliar situations	Ability to cultivate sincerity, honesty, fairness, and respect creativity	Creativity
Ability to find quality information in the flood of information	Ability to listen attentively	Ability to build trust, resolve conflicts, and support others	Ability to utilize multimedia resources to effectively communicate one's ideas in a variety of formats	Ability to use one or more of the international languages used worldwide	Ability to explore and experiment	Ability to display moral courage in the face of injustice	A love of learning and curiosity
Ability to articulate meaningful questions	Ability to speak effectively and coherently	Ability to facilitate group discussions or discussions, build consensus, and negotiate outcomes	Ability to master and use higher mathematics	Ability to use technology to connect people and things around the world	Ability to word effectively in situations of ambiguous or changing priorities	Ability to act responsibly for the benefit and happiness of the greater community	Resilience/elasti city
Ability to analyze and create ideas or knowledge	Ability to write concisely and clearly, considering the diversity of readers	Ability to teach, guide, and counsel Ability to seek help from others	Ability to understand traditional or recently emerging fields such as mathematics, science, technology, environmental science, robotics, geometry, cellular automata, nanotechnology, and	Social and intellectual skills to effectively navigate information across multiple cultures	Ability to see failure as a lesson and to accept that innovation is made up of frequent failures and small successes	Ability to have an intrinsic understanding of ethical issues or dilemmas about new media or new technologies	Perseverance/pa tience

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			biotechnology				
Ability to devise and test solutions to problems by trial and error	Ability to explain information and persuasively engage others in an engaging way	Ability to collaborate on tasks, manage groups, and delegate responsibilities		Ability to use 21 st century core competencies such as communication skills, collaboration skills, critical thinking skills, creativity to properly understand and deal with global issues	Ability to develop a sense of independence to seek new roles, ideas, or strategies	Ability to make rational and ethical decisions on complex issues	Self-efficacy
Ability to think of different ways to cope		Ability to listen and practice goals		Ability to learn from and collaborate with people from different cultures, religions and lifestyle through mutual respect and open dialogue	Ability to develop the knowledge and competencies necessary to start a business and run a business		Stress management skills
Ability to develop knowledge and perspectives that are not limited to one field but span multiple fields		Ability to share credit for success		Ability to generate diverse ideas and exploit social and cultural differences to get things done			Time management skills
Ability to make sustained inferences							
Ability to put things together or adapt them to new uses and situations							
Ability to solve new problems that cannot be solved through established rules							
Ability to use knowledge or creativity to solve complex							

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real-world problems							

1. Complex Problem-Solving Skills	Ability to solve new and unclear problems that may arise in complex real-world situations
2. Critical Thinking Skills	Ability to compare and analyze the strengths and weaknesses of various solutions or approaches to a given problem using logic and reasoning
3. Creativity	Ability to come up with novel ideas when given a certain topic or situation or to suggest a creative method in the process of solving a problem
4. Human Resource Management Skills	Ability to select the most suitable person for a specific task and manage them to achieve maximum work performance through motivation, and systematic efficient management
5. Ability to Collaborate	Ability to adjust one's own behavior in response to the behavior of others
6. Emotional Ability	Ability to perceive other people's reactions and understand why they react the way they do
7. Judgement and Decision-making Skills	Ability to make the most rational judgement and decision by analyzing the costs and benefits of a specific situation or problem
8. Orientated Towards Service	Ability to actively seek ways to help others
9. Negotiation Skills	Ability to bring together parties who differ in ideas or interests and try to reduce those differences
10. Cognitive Flexibility	Ability to apply different kinds of rules or principles to rearrange and reorganize things in different ways